

Omnimodal Models

Gunhee Kim



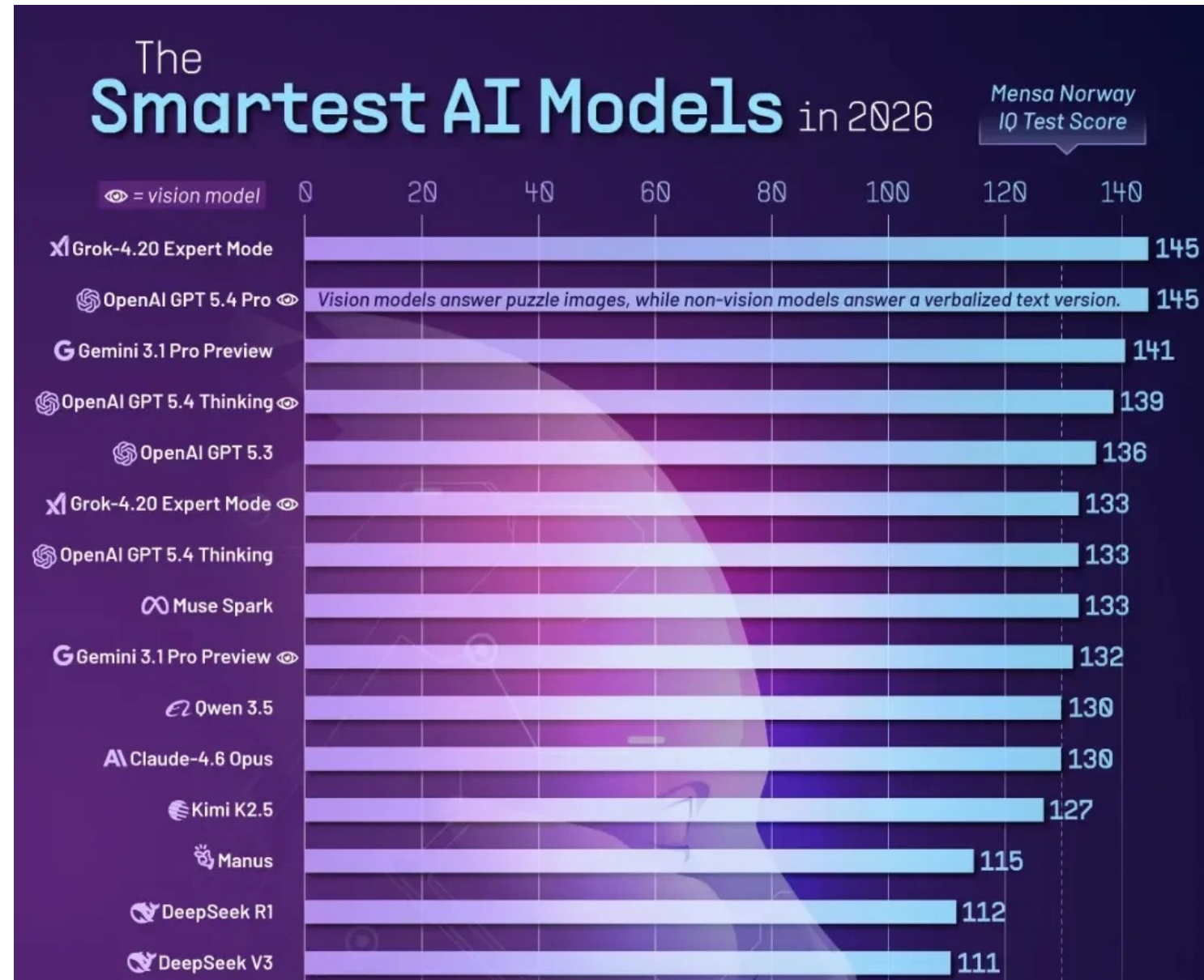
서울대학교
SEOUL NATIONAL UNIVERSITY

Contents

- Foundation Models
- Omnimodal Models
- Robot Foundation Models

Smartest AI Models in 2026

- 2026 Mensa Norway benchmark (2026 April)
 - The top tier is getting crowded, with several leading models now separated by only a few points
 - 130 Genius, 100 Avg human



<https://www.visualcapitalist.com/ranked-the-smartest-ai-models-in-2026/>

Emergence of Frontier Models

- A general-purpose foundation model to perform a wide variety of tasks across multiple domains
 - Trained from extremely large computational budgets on the order of 10^{26} FLOPS
 - Multimodal frontier: perceive and reason across video, audio, and text, and understand the physical world as fluently as they understand language

Scale and complexity	General-purpose capabilities	Emergent behaviors	Societal impact
Extremely large models trained with massive datasets	Perform a wide range of distinct tasks	Unexpected skills arose from scale and training	Can transform industries and raises ethical questions

FLOP: floating-point operations per second

Open Source vs Closed Source Models

Set the upper bound for reasoning and general intelligence



GPT-5

Industry-leading reasoning, tool use, and developer ecosystem



Claude opus-4

Long-context reasoning, safety-first, B2B applications



Gemini 3

Tight integration with Google's ecosystem



Grok 4

Real-time knowledge and social-context awareness

Increasingly challenge proprietary dominance



Llama 4

Sovereignty-friendly, high-performance general model



Large 3

Efficiency-focused frontier reasoning



Qwen3

Model family with strong multilingual and multimodal capabilities



Deep Seek-4

Exceptional price-performance trade-off

Open Source vs Closed Source Tradeoff

- Performance vs cost
 - Proprietary models: (1) more expensive at inference time, (2) Subject to usage limits and pricing changes, (3) opaque in terms of training data and model
 - Open-weight models deliver 80–95% of flagship performance at a fraction of cost
 - No longer just about **maximum intelligence**, but about **intelligence per dollar**

Model	Input	Output	Context Window
OpenAI GPT-5.5	\$5.00	\$30.00	1M
OpenAI GPT-4.1	\$2.00	\$8.00	1M
Claude Opus 4.7	\$5.00	\$25.00	1M
Llama 4 Maverick	\$0.20	\$0.60	1M
Mistral Large 3	\$0.50	\$1.5	256K
DeepSeek-4	\$0.14	\$0.28	1M

Input/output price
per 1M tokens

All models have
cheaper options

All prices are
subject to change

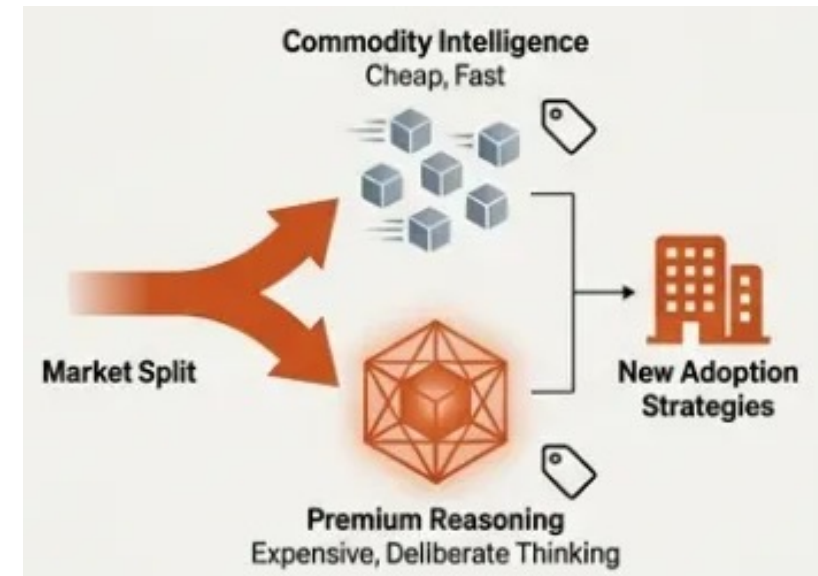
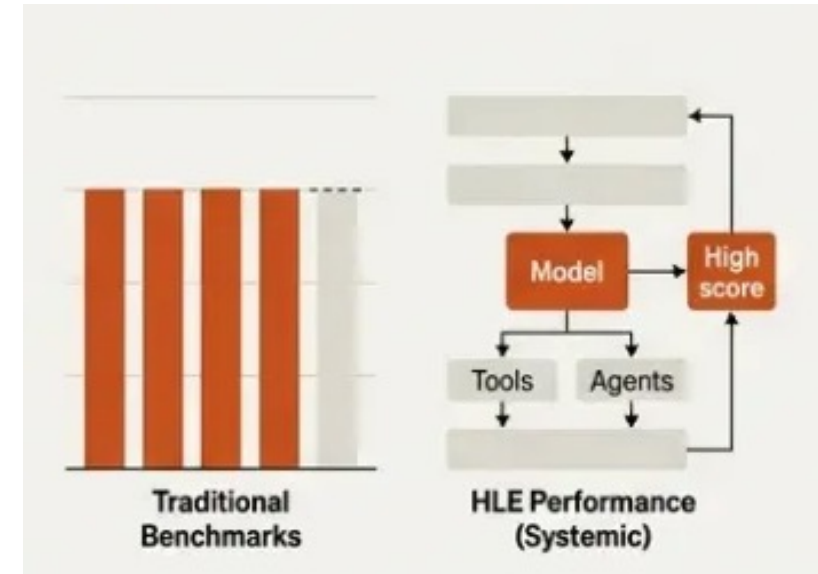
The Age of Orchestration

- The era of the standalone LLM is over
 - Orchestrated systems combine models with tools, agents, and multi-step reasoning
 - System 2 agentic reasoning, inference-time compute, multi-agent collaboration
 - Simply training a bigger model on more data no longer guarantees leadership



The Age of Orchestration

- Performance is now systematic
 - Traditional benchmarks are saturated
 - Peak performance is not by the model alone, but by the system orchestrating it with tools and agents
- Economic bifurcation
 - New enterprise adoption strategies: commodity intelligence (cheap, fast model) + premium reasoning (expensive, deliberate model)



Openrouter's Leaderboard

- Chinese AI models hit 61% market share on OpenRouter
 - AI API gateway lets developers access hundreds of LLMs from many providers through a single API
 - Heavy in agentic flows run by U.S. firms (Roughly 80% of startups use Chinese models)



LLM Leaderboard

Compare the most popular models on OpenRouter ⓘ

All models

This Week

1.		Hy3 (free) by tencent	6.13T tokens new	6.		DeepSeek V4 Pro by deepseek	2.83T tokens ↑ 20%
2.		MiMo - V2.5 by xiaomi	5.95T tokens ↑ 36%	7.		Claude Opus 4.7 by anthropic	2.26T tokens ↑ 3%
3.		DeepSeek V4 Flash by deepseek	5.22T tokens ↓ 2%	8.		Nemotron 3 Ultra (free) by nvidia	2.04T tokens ↑ 109%
4.		MiniMax M3 by minimax	4.26T tokens ↑ 4%	9.		Claude Opus 4.8 by anthropic	2.03T tokens ↓ 3%
5.		GLM 5.2 by z-ai	3.19T tokens ↑ 24%	10.		Step 3.7 Flash by stepfun	1.22T tokens ↓ 21%

Code Arena's Votes



- Moonshot AI's brand-new model is comparable to Claude Fable 5
 - AI evaluation platform to test how LLMs plan, code, iterate, and build real-world software applications
- Rankings on front-end web development tasks

Code Arena | WebDev

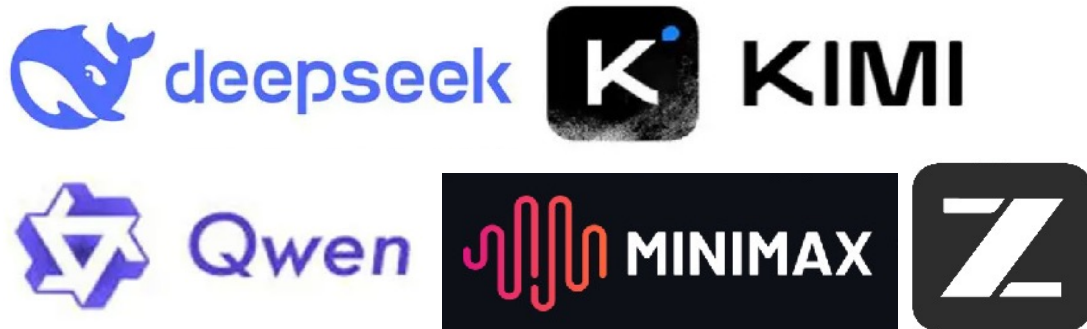
Overall

Rank	Rank Spread	Model	Score	Votes
1	1 2	claude-opus-5-max Anthropic · Proprietary	1712 +20/-20 Preliminary	1,278
2	1 3	kimi-k3-max Moonshot · Kimi K3 license	1682 +13/-13 Preliminary	3,777
3	2 3	claude-opus-5-high Anthropic · Proprietary	1669 +13/-13	2,855
4	4 5	claude-fable-5 Anthropic · Proprietary	1628 +10/-10	5,887
5	4 5	gpt-5.6-sol-xhigh (codex-h... OpenAI · Proprietary	1623 +10/-10	5,460
6	6 6	glm-5.2-max Z.ai · MIT	1588 +9/-9	5,865

<https://arena.ai/leaderboard/code/webdev/overall>

Who Will Win the AI Race: The U.S. or China?

- The U.S. has power constraint vs. China has silicon constraint
 - Recently, Z.ai built 1GW data center using only China-made chips
 - Huawei's Ascend chips consume x4 power as NVIDIA chips for comparable computing performance
 - However, electricity is much cheaper in China than in U.S, what if it is 4x cheaper?



No Moat!

- Internal Google memo, May 2023
 - *But the uncomfortable truth is, we aren't positioned to win this arms race and neither is OpenAI*
- DeepSeek shock
 - Founded in May 2023, and release DeepSeek-R1, comparable to OpenAI O1 in December 2024



The most important thing [is] trying to find a business with a wide and long-lasting moat around it - 1995 Annual Meeting

Google "We Have No Moat, And Neither Does OpenAI"

Leaked Internal Google Document Claims Open Source AI Will Outcompete Google and OpenAI

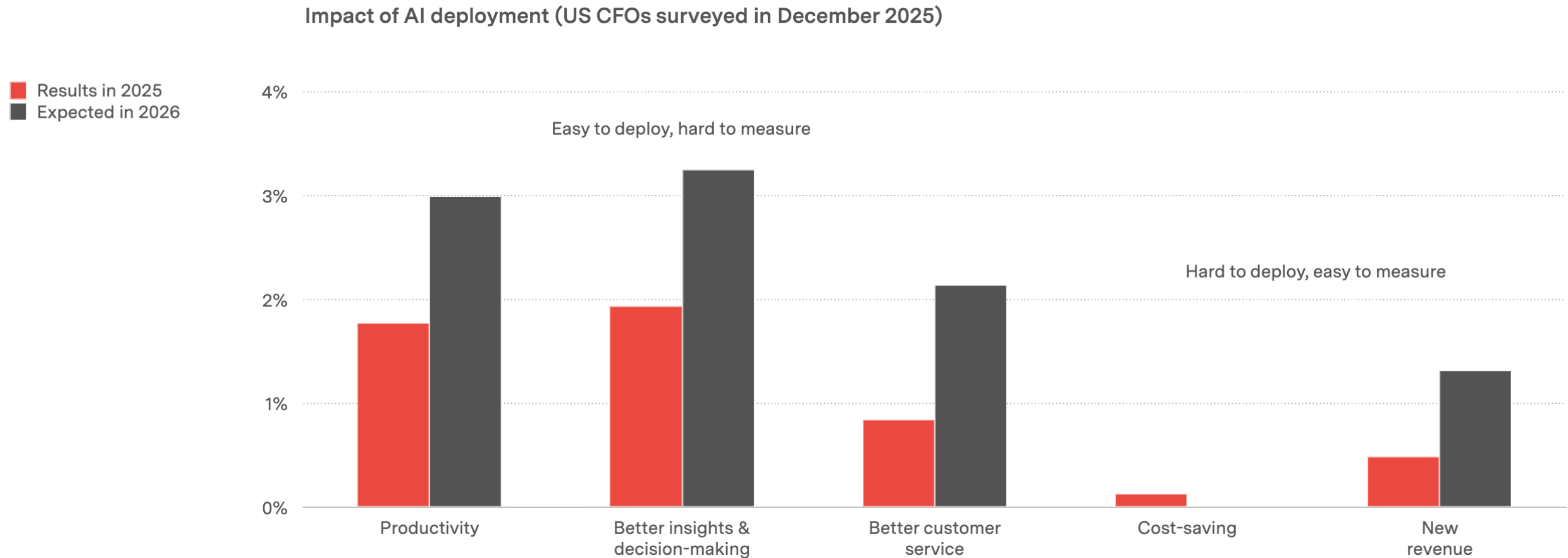
DYLAN PATEL AND AFZAL AHMAD

MAY 04, 2023 • PAID



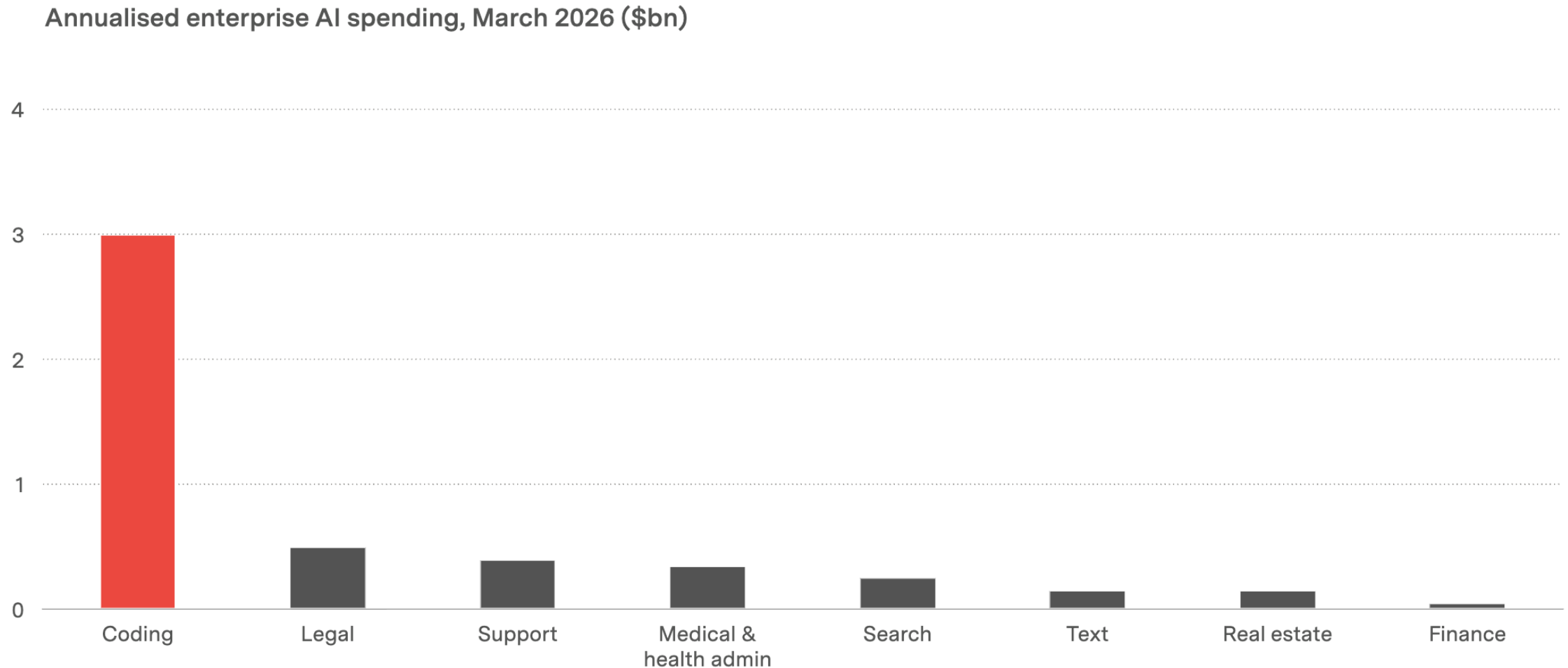
What Works first?

- First, automate the obvious use cases - innovation takes longer



Product Market Fit

- A generational change in software development



Important Quotes



In my lifetime, I've seen two demonstrations of tech that struck me as revolutionary: the GUI and ChatGPT



The risk of under-investing is significantly greater than the risk of over-investing



The very worst case would be that we have just pre-built for a couple of years

Summary

- Choosing the right model
 - Need the absolute best reasoning available → closed models
 - Need cost efficiency, control, and customization → open-weight models
 - A hybrid approach is often the most resilient long-term strategy
- Challenges
 - Alignment and fairness: Reliable, predictable, and consistent with human intent
 - Safety: Scalable misinformation and deepfakes, automated social engineering and phishing, malware generation or vulnerability discovery
 - Sustainability: Massive GPU or accelerator clusters, enormous energy consumption, and significant water usage for data-center cooling

Contents

- Foundation Models
- **Omnimodal Models**
- Robot Foundation Models

OpenAI GPT-4 (5/14/2023)

- One of the first multimodal language model
 - Accept image and text inputs, and generate text outputs
 - OpenAI GPT-4V(ision) is available on on November 6th, 2023

Scene text recognition



Q) What are all the scene text in the image?

A) The only scene text in the image is "The Earth without art is just 'eh'".

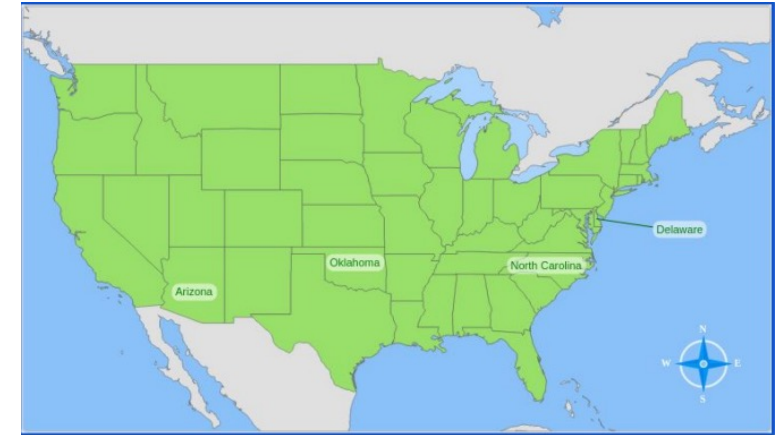
Landmark recognition



Q) Describe the landmark in the image.

A) The landmark in the image is the statue of George M. Cohan in Times Square, New York City. It is ...

Science and knowledge

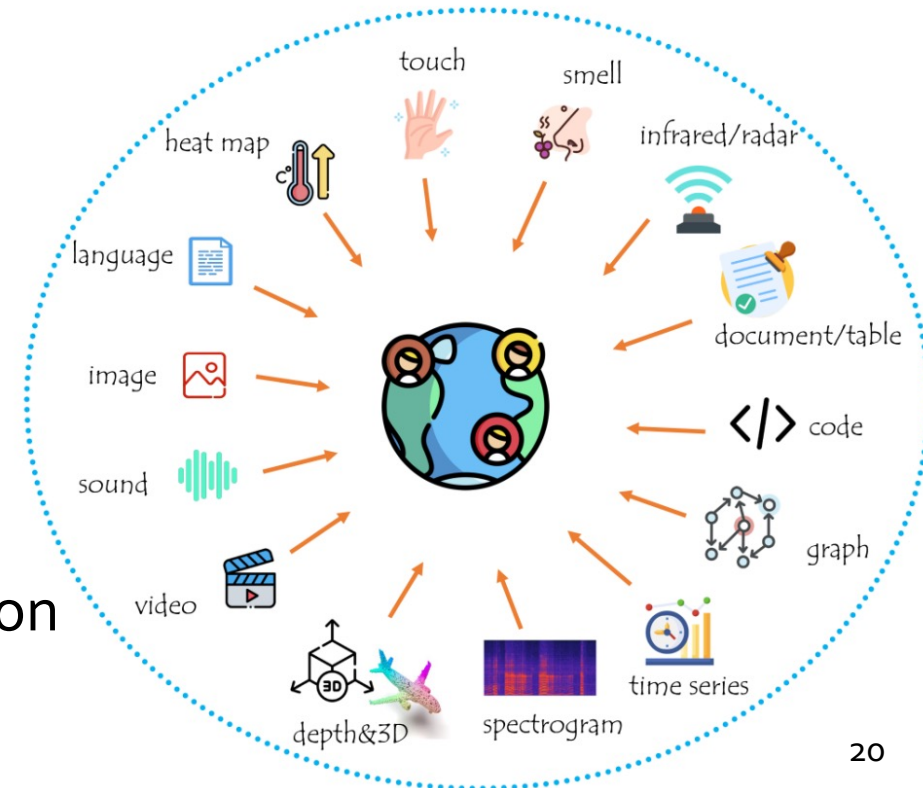


Q) Which of these states is farthest north?

A) Out of the three states shown in the map - Arizona, Oklahoma, and Delaware, Delaware is the farthest north.

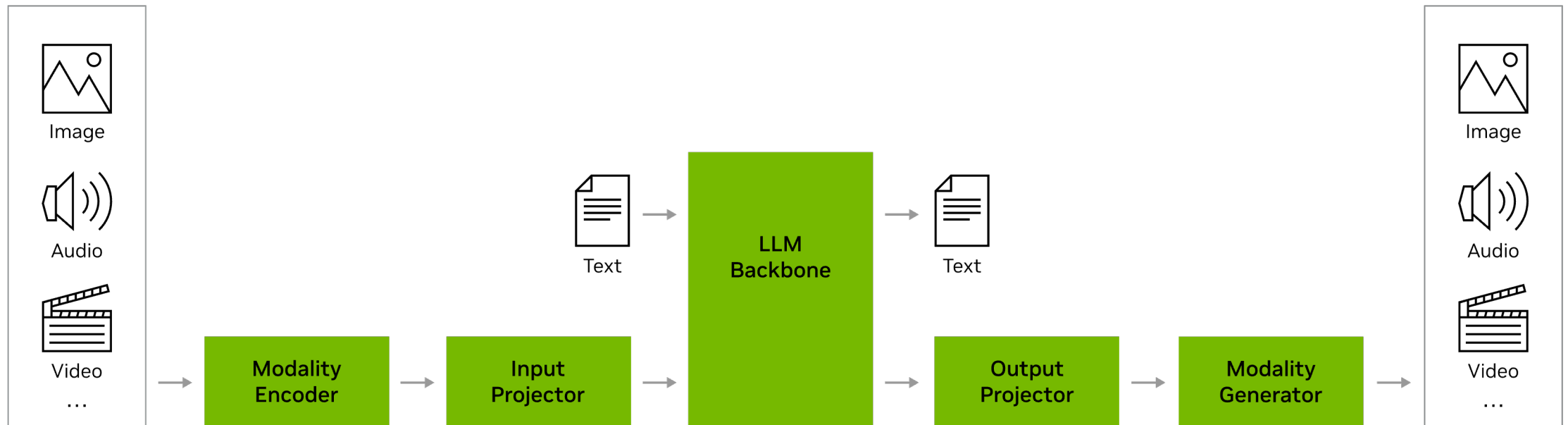
What is Multimodality?

- Definition of modality
 - Modality refers to the way in which something expressed or perceived
 - Multimodal research studies **heterogeneous** and **interconnected** data
 - e.g., a video: frames + speech + sounds ...
- Our experience of the world is multimodal
 - This world we live in is replete with multimodal information and signals, **not just language**
 - We see objects, hear sounds, feel the texture, smell odors, taste flavors, ...
 - Human acquires 70% of daily information from vision



Build Multimodal LLMs (MLLMs)

- LLMs are already (almost) AGI and evolving into multimodal LLMs
 - AGI matches or exceeds human intelligence; it does NOT just simulate humans, but they are intelligent on their own
 - Human use language as the common space to share information
 - *“The limits of my language mean the limits of my world.”* — Ludwig Wittgenstein



GPT-4O(mni) (5/13/2024)

- More natural human-computer interaction
 - A single neural network model trained end-to-end across all inputs and outputs
 - Input/output: any combination of text, audio, image, and video inputs
 - Omni-modal capabilities: derived from the Latin word *omnis*, meaning all or every
 - Response time: 232 ms (c.f. 320 ms of human conversation)
 - Distinguish different breathing patterns, speakers, emotions, and volume and pacing of speech
 - Multilinguality: 50 languages over 97% of speakers



<https://openai.com/index/hello-gpt-4o/>

<https://youtu.be/V6pYxfcDRks>

GPT-5 (8/7/2025)



Over 2 years ago, Bill Gates claimed that OpenAI's technology had **reached a plateau**, indicating that GPT-5 wouldn't be significantly better than GPT-4

Sora 2 Is Shutting Down

- Video generation AI by OpenAI
 - Create highly realistic, consistent, and long-form video clips (and audio) from simple text or image prompts
 - Shutting down Sora web and app experiences on April 26, 2026, followed by the API on September 24, 2026
 - Expensive to run, with low monetization
 - *Focus on world simulation research to advance robotics that will help people solve real-world, physical tasks*



Sora 2 Is Shutting Down

- OpenAI is in big trouble
 - Promised adult mode - now shelved
 - Launched Sora video generator, Disney's \$1 billion equity investment in OpenAI - ended Sora 100 days later
 - Announced Stargate project - cancelled one year later
 - Altman once called AI + ads a "last resort" - 16 months later launched ads
 - Launched in-app shopping with direct checkout - now cancelled
 - Promised first hardware device this year - now delayed to 2027

Newsletters

The Atlantic

TECHNOLOGY

OpenAI Is Doing Everything ... Poorly

The company's sudden decision to pull the plug on Sora is a sign of deeper trouble.

Contents

- Foundation Models
- Omnimodal Models
- Physical AI

Figure AI

- AI-powered humanoid robots
 - *This week, Figure has passed 5 months running on the BMW X3 body shop production line. We have been running 10 hours per day, every single day of production! It is believed that Figure and BMW are the first in the world to do this with humanoid robots. (2025.10.06)*
– CEO Brett Adcock

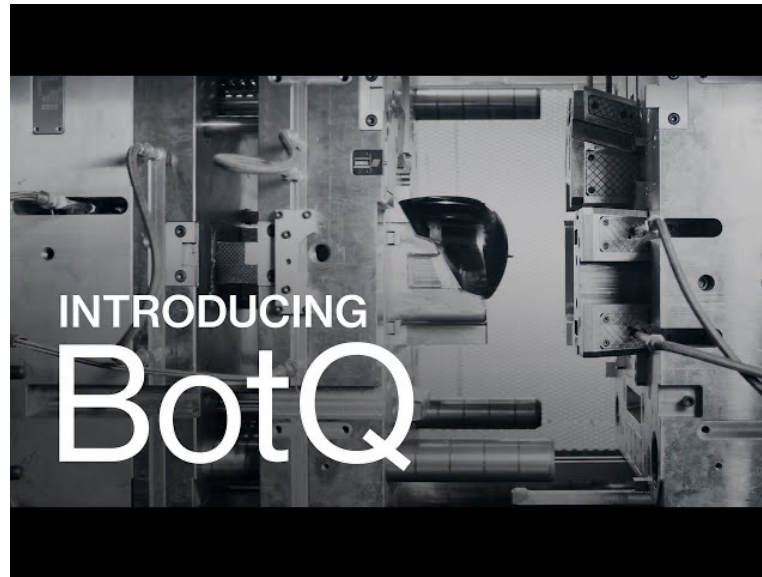


https://www.reddit.com/r/singularity/comments/1nzm79t/brett_adcock_this_week_figure_has_passed_5_months/



Figure AI

- Figure 03: a third-generation humanoid (2025.10.9)
 - Designed for home and factory
 - A manufacturing facility BotQ will produce 12,000 Figure 03 humanoids per year (2025.03.15)



Tesla Optimus

- A general-purpose robotic humanoid
 - *Optimus has the potential to be more significant than [Tesla's] vehicle business over time* – Elon Musk (January 2022)
 - Why \$1 trillion pay package for Musk? *\$3 trillion by the end of 2026 as full scale volume production begins of the autonomous and robotics roadmap* (2025.11.7)



Boston Dynamics (Owned by Hyundai Motors Group)

- Shifting from R&D to commercialization
 - Atlas: earliest commercialized bipedal humanoid robot in 2016
 - Focus on selling high-volume robots for industrial and commercial applications like industrial monitoring of facilities (Spot, most used mobile autonomous robots over 500 customers over 46 countries) and warehouse automation (Stretch)



Unitree Robotics



- Pioneering affordability
 - Founded by Wang Xingxing (1990) in May 2016
 - The H2 humanoid robot is priced around \$29,000
 - Unitree Go2 can be purchased in Coupang below 3M KRW



What is Physical AI?

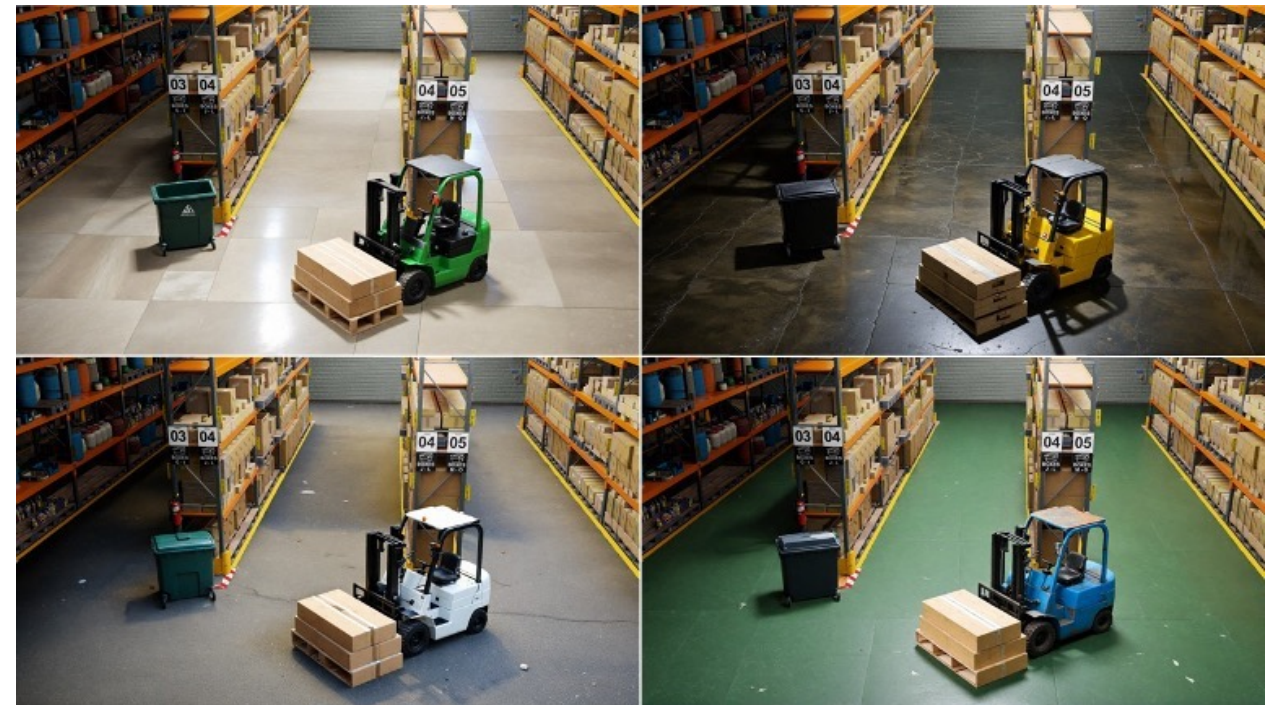
- AI should perceive, understand, and interact with the physical world by using sensors and actuators to gather data and take actions
 - cf. Embodied AI refers to the integration of AI into physical systems, enabling them to interact with the physical world
 - e.g., robots, autonomous vehicles, smart spaces like factories and warehouses, and any network with sensors and actuators in hospital or home scenarios
 - One intrinsic feature is the uncertainty!
 - The uncertainty associated with the acquired information, and incompleteness and uncertainty about the effects of actions over (physical) systems that share the environment with humans

Data Scarcity

- AI must handle diverse, complex situations
 - Autonomous driving: construction zones, weather, parked vehicles
 - Warehouse robotics: lighting, weight, materials, layout



Different types of driving conditions



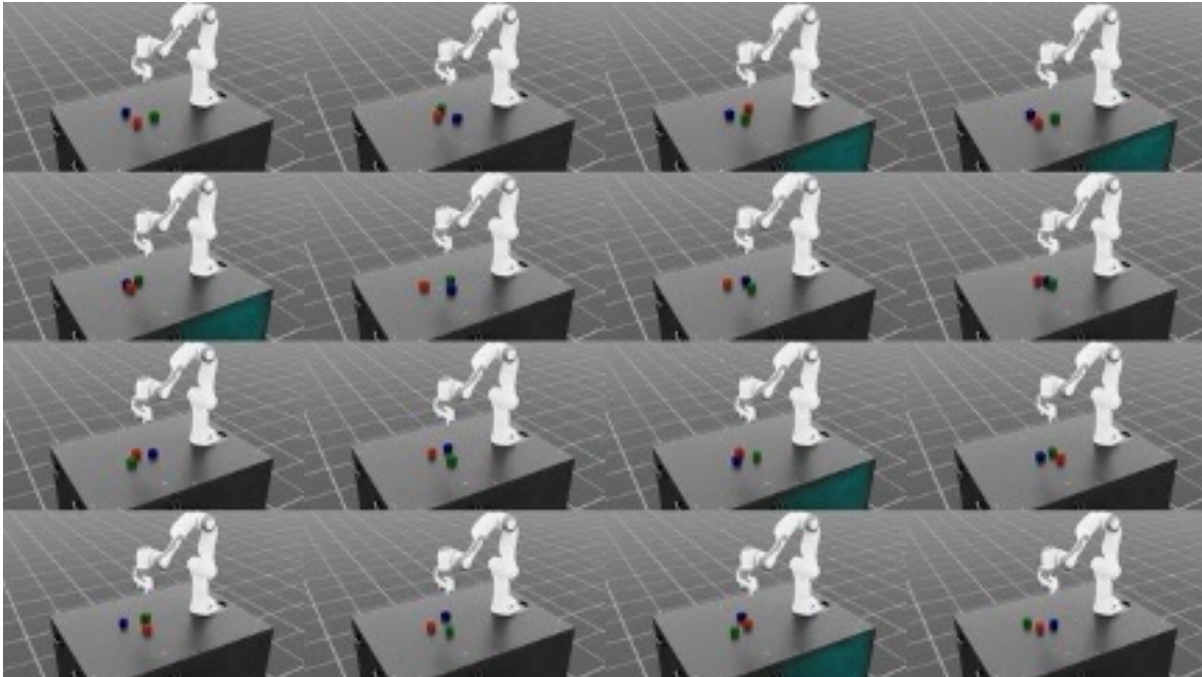
Different types of flooring and forklift colors

NVIDIA Physics AI Ecosystem

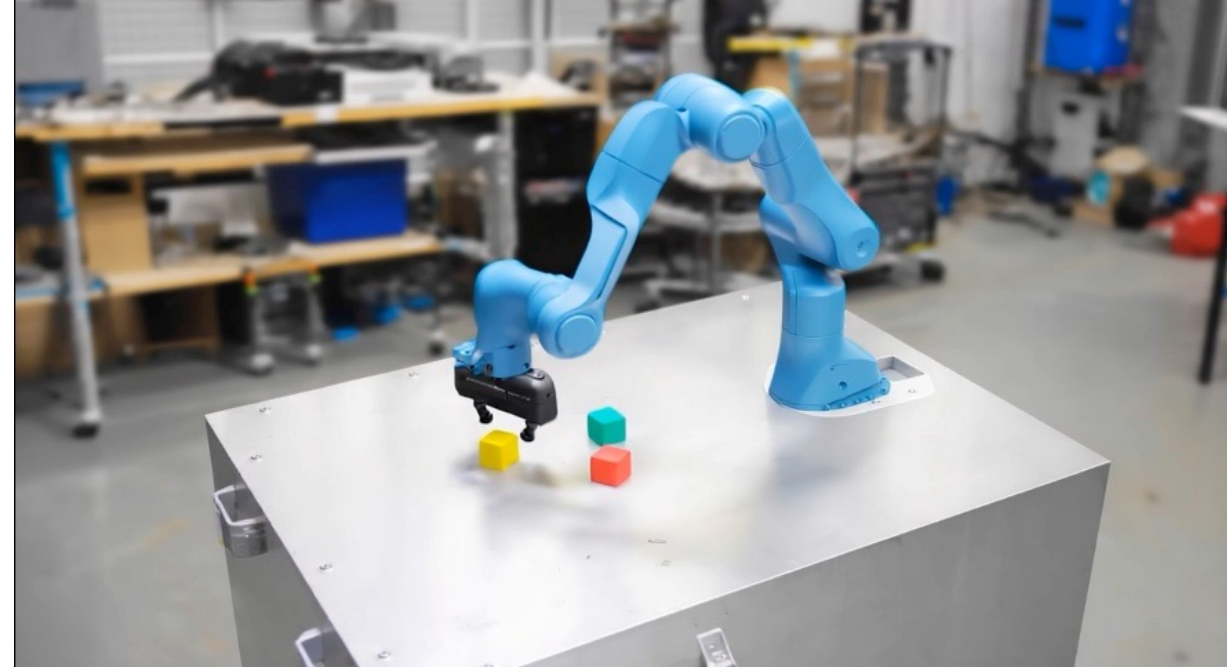
- Omniverse: NVIDIA's simulation engine
 - Photorealistic, physically accurate virtual environments
 - Simulates real-world physics: light reflection & shadows, object collision & gravity
 - Edify (3D asset generation): generate realistic, high-quality 3D content for virtual environments
- Cosmos: world foundation model
 - Trained to understand real-world dynamics across diverse domains
- GRooT: robot foundation model
 - Designed for physical interaction
 - Built on large-scale synthetic trajectory data generated via Cosmos

NVIDIA Omniverse

- A real-time 3D graphics collaboration platform
 - Physics-aware data for training models that generalize well
 - User can vary robot arm material, background environment, and material of boxes



Generate various object placements



Synthetically generated videos

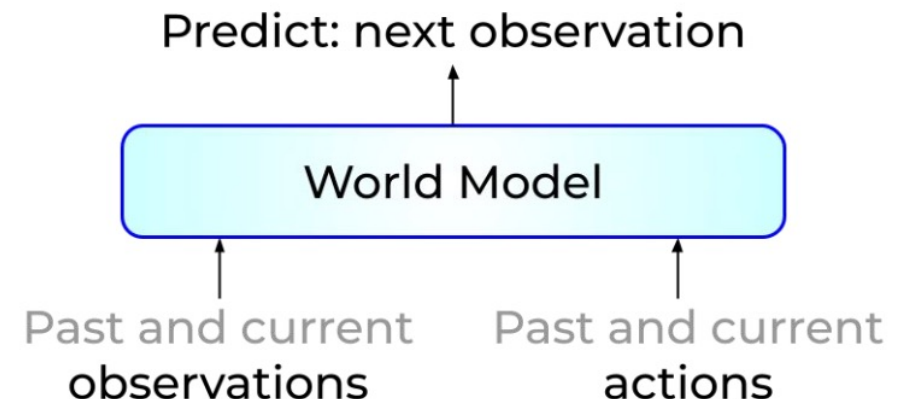
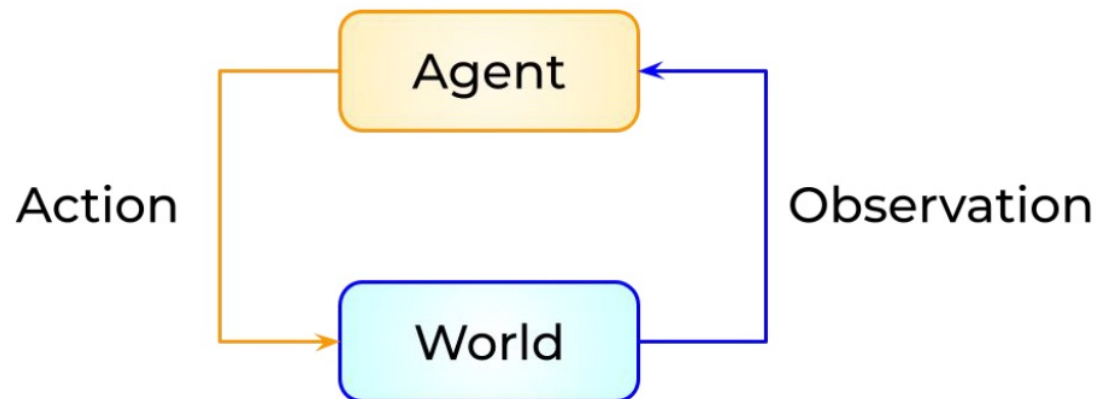
NVIDIA Cosmos – Autonomous Driving

- Driving datasets can be fully generated in simulation
 - Create rich, scenario-specific training data by digitally twinning the real world,
 - Learn not just basic navigation, but also how to handle rare or complex events such as construction zones, blocked signals, or sudden obstacles



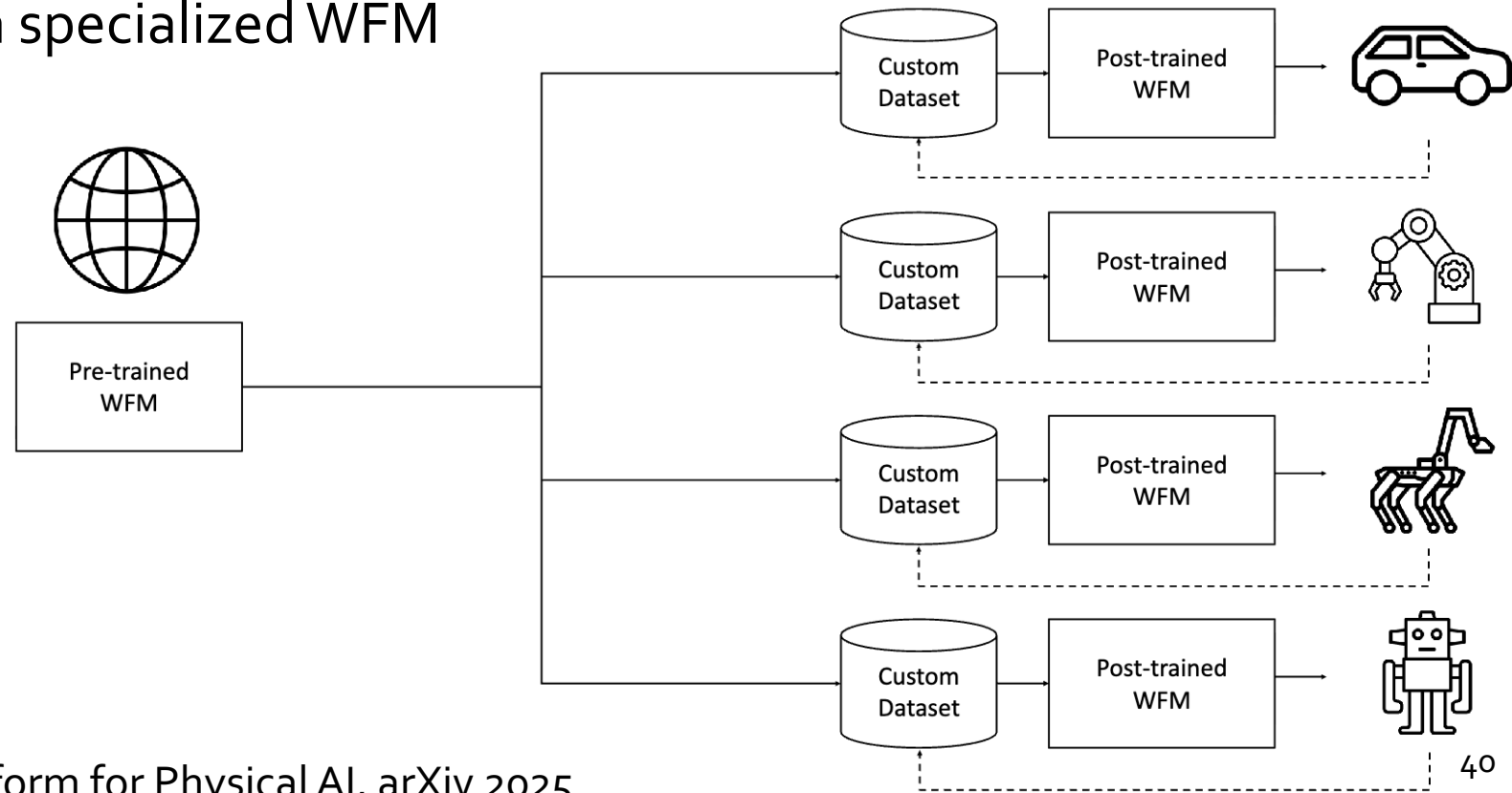
Definition of WFMs

- World models predict the next observation in an environment given the current action and the past observations
 - Reinforcement learning (RL): a machine learning approach where an AI agent learns by interacting with an environment and receiving rewards or penalties based on its actions
 - Over time, it optimizes decision-making to achieve the best possible outcome



NVIDIA Cosmos

- A pre-training and-then-post-training paradigm
 - Pretraining: a large-scale video training dataset to expose the model to a diverse set of visual experiences to become a generalist
 - Post-training: fine-tune a specialized WFM using a dataset collected for the targeted, specialized physical AI setup



Other World Model – World Labs

- Marble – persistent, navigable, and controllable 3D worlds
 - Given an image or text prompt, generate a 3D world that one can explore — no time limits, no morphing, no inconsistency
 - Better geometry and more diverse styles

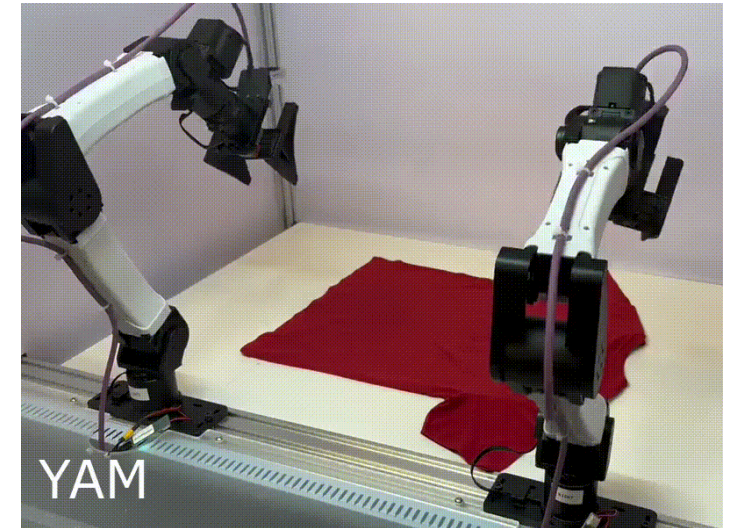


Robotic Foundation Models

- Large-scale AI models designed to serve as versatile building blocks for a wide array of robotics tasks
 - One of Vision-Language-Action (VLA) models
 - Like LLMs for robots (generalization across diverse robotic applications by extensive training on vast datasets)
 - Unlike traditional model for specific tasks like object manipulation or navigation, foundation models are pre-trained on diverse datasets that encompass a wide range of robotic scenarios
 - Multimodal learning: integrate data from various sensors (e.g., cameras, lidar, and tactile sensors) to understand and respond to their environment holistically
 - Generalization: perform well in tasks they were not explicitly trained on
 - Fine-Tuning: adapt to specific tasks with minimal additional training

NVIDIA Isaac GRooT

- Open foundation model for generalized humanoid robot skills
 - NVIDIA Isaac: AI robot development platform (libraries, application frameworks, and AI models)
 - Cross-embodiment model takes multimodal input (language and images, to perform manipulation tasks in diverse environments

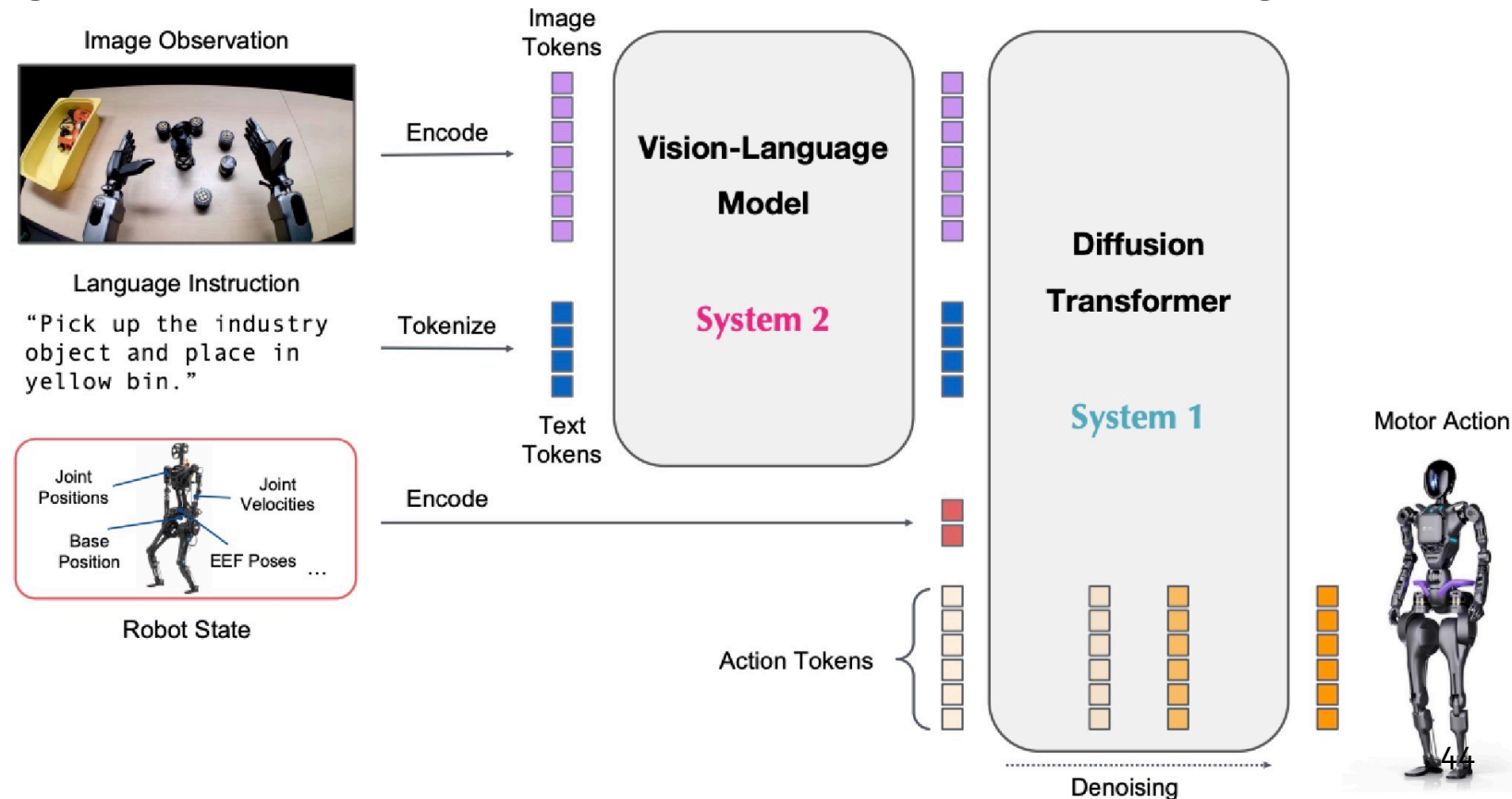


<https://github.com/NVIDIA/Isaac-GRooT>

NVIDIA, GRooT N1: An Open Foundation Model for Generalist Humanoid Robots, arXiv 2025

NVIDIA Isaac GRooT

- A dual-system compositional architecture
 - System 1: a fast-thinking action model, mirroring human reflexes or intuition
 - System 2: a slow-thinking model for deliberate, methodical decision-making
- Inspired by human cognitive processing



Summary

- Potential challenges
 - Generalization: robots/tasks are too diverse!
 - Robust real-world deployment: robots should perform reliably in diverse and unpredictable environments
 - Data collection: real robot data should be gathered for each specific robot
 - Recovery from failure: the dataset often contains only successful demos
 - Sequence length limit: this may not solve long horizon tasks
 - Collaborative learning: sharing datasets and models across organizations can accelerate progress
 - Human-robot interaction: how to understand and respond to human cues will be critical for widespread adoption